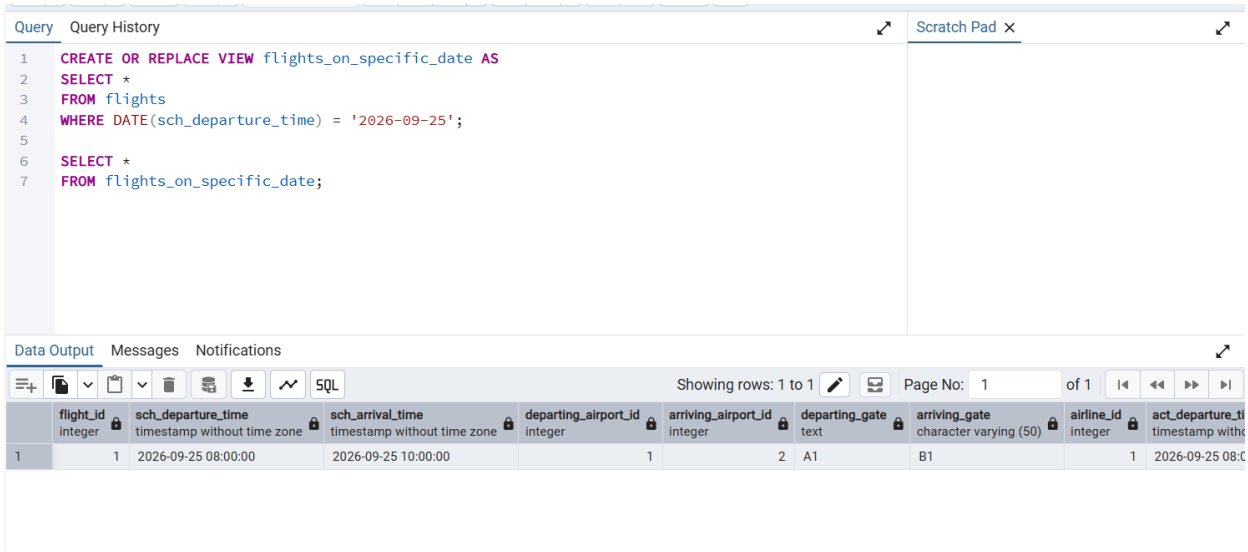
LAB WORK №8

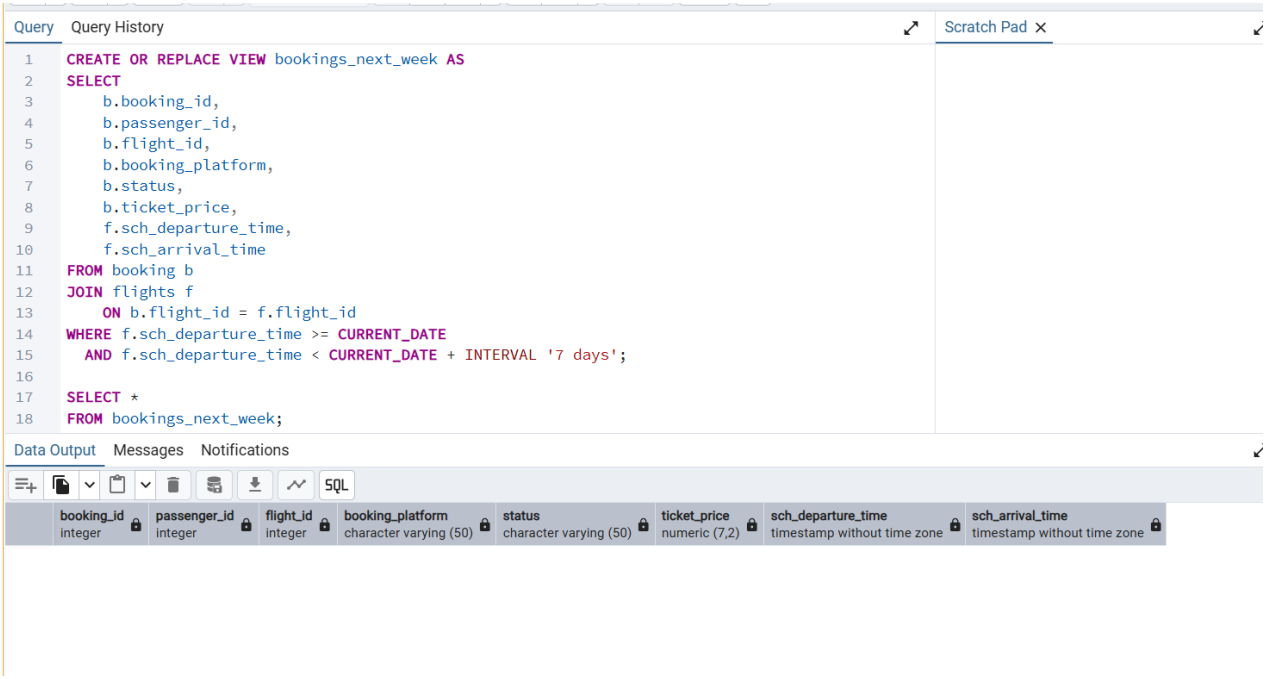
VIEW

Задание 1. Создать представление рейсов на определённую дату



image(20261110-011533).png

Задание 2.



image(20261110-011615).png

Задание 3.

Query | Query History

```sql
CREATE OR REPLACE VIEW top_5_popular_routes AS
SELECT
    dep.airport_name AS departure_airport,
    arr.airport_name AS arrival_airport,
    COUNT(b.booking_id) AS total_bookings
FROM booking b
JOIN flights f
    ON b.flight_id = f.flight_id
JOIN airport dep
    ON f.departing_airport_id = dep.airport_id
JOIN airport arr
    ON f.arriving_airport_id = arr.airport_id
GROUP BY
    dep.airport_name,
    arr.airport_name
ORDER BY total_bookings DESC
LIMIT 5;

SELECT *
FROM top_5_popular_routes;
```

Data Output | Messages | Notifications

Showing rows: 1 to 5

	departure_airport character varying (50)	arrival_airport character varying (50)	total_bookings bigint
1	Almaty International Airp...	Astana International Airp...	2
2	Frankfurt Airport	Almaty International Airp...	1
3	Istanbul Airport	Dubai International Airport	1
4	Astana International Airp...	Istanbul Airport	1
5	Dubai International Airport	Frankfurt Airport	1

image(20261110-011647).png

Задание 4.

```sql
SELECT
    airline_id,
    airline_name,
    airline_code,
    airline_country
FROM airline
ORDER BY airline_id;
```

	airline_id [PK] integer	airline_name character varying (50)	airline_code character varying (30)	airline_country character varying (50)
1	1	Air Astana	KC	KAZAKHSTAN
2	2	FlyArystan	FS	KAZAKHSTAN
3	3	Turkish Airlines	TK	TURKEY
4	4	Emirates	EK	UNITED ARAB EMIRAT...
5	5	Global Airways	LH	United Kingdom
6	6	KazAir	KZ	TURKEY
7	7	AirEasy	AE	FRANCE
8	8	Dana	FH	BRAZIL
9	9	FlyFly	FF	POLAND
10	10	wd	aa	usze

image(20261110-011724).png

```sql
SELECT
    f.*,
    a.airline_name,
    a.airline_code
FROM flights f
JOIN airline a
    ON f.airline_id = a.airline_id
WHERE a.airline_id = 1;

SELECT *
FROM flights_by_airline;
```

	flight_id integer	sch_departure_time timestamp without time zone	sch_arrival_time timestamp without time zone	departing_airport_id integer	arriving_airport_id integer	departing_gate text	arriving_gate character varying (50)	airline_id integer	act_departure_ti timestamp witho
1	1	2026-09-25 08:00:00	2026-09-25 10:00:00	1	2	A1	B1	1	2026-09-25 08:0

image(20261110-011805).png

Query | Query History

```
CREATE OR REPLACE VIEW flights_by_airline AS
SELECT
    f.*,
    a.airline_name,
    a.airline_code
FROM flights f
JOIN airline a
    ON f.airline_id = a.airline_id
WHERE a.airline_id = 1
  AND f.sch_departure_time >= CURRENT_DATE
  AND f.sch_departure_time < CURRENT_DATE + INTERVAL '7 days';

SELECT *
FROM flights_by_airline;
```

Scratch Pad

Data Output | Messages | Notifications

flight_id integer	sch_departure_time timestamp without time zone	sch_arrival_time timestamp without time zone	departing_airport_id integer	arriving_airport_id integer	departing_gate text	arriving_gate character varying (50)	airline_id integer	act_departure_ti timestamp witho

image(20261110-011833).png

Задание 5.

Query | Query History

```
SELECT
    f.*,
    a.airline_name,
    a.airline_code
FROM flights f
JOIN airline a
    ON f.airline_id = a.airline_id
WHERE a.airline_id = 1;

SELECT *
FROM flights_by_airline;
```

Scratch Pad

Data Output | Messages | Notifications

Showing rows: 1 to 1 | Page No: 1 of 1

	flight_id integer	sch_departure_time timestamp without time zone	sch_arrival_time timestamp without time zone	departing_airport_id integer	arriving_airport_id integer	departing_gate text	arriving_gate character varying (50)	airline_id integer	act_departure_ti timestamp witho
1	1	2026-09-25 08:00:00	2026-09-25 10:00:00	1	2	A1	B1	1	2026-09-25 08:0

image(20261110-011805).png

Задание 6.

Query | Query History

```
1  CREATE OR REPLACE VIEW flights_delayed_over_24_hours AS
2  SELECT
3      flight_id,
4      sch_departure_time,
5      act_departure_time,
6      act_departure_time - sch_departure_time AS delay_duration
7  FROM flights
8  WHERE act_departure_time IS NOT NULL
9    AND act_departure_time > sch_departure_time + INTERVAL '24 hours';
10
11 SELECT *
12 FROM flights_delayed_over_24_hours;
```

Data Output | Messages | Notifications

	flight_id integer	sch_departure_time timestamp without time zone	act_departure_time timestamp without time zone	delay_duration interval

image(20261110-011904).png

Задание 7.

Query | Query History

```sql
CREATE OR REPLACE VIEW leffler_thompson_passengers AS
SELECT DISTINCT
    p.first_name || ' ' || p.last_name AS full_name,
    p.country_of_citizenship AS country
FROM passengers p
JOIN booking b
    ON p.id1 = b.passenger_id
WHERE b.booking_platform = 'Leffler-Thompson';

SELECT *
FROM leffler_thompson_passengers;
```

Data Output | Messages | Notifications

full_name text	country character varying (50)

image(20261110-012004).png

Задание 8.

Query | Query History

```
1  CREATE OR REPLACE VIEW top_10_visited_countries AS
2  SELECT
3      a.country,
4      COUNT(b.booking_id) AS total_visits
5  FROM booking b
6  JOIN flights f
7      ON b.flight_id = f.flight_id
8  JOIN airport a
9      ON f.arriving_airport_id = a.airport_id
10 GROUP BY a.country
11 ORDER BY total_visits DESC
12 LIMIT 10;
13
14 SELECT *
15 FROM top_10_visited_countries;
```

Data Output | Messages | Notifications

Showing rows: 1 to 4

	country character varying (50)	total_visits bigint
1	Kazakhstan	3
2	Germany	1
3	United Arab Emirates	1
4	Turkey	1

image(20261110-012035).png

Задание 9.

Query | Query History | Scratch Pad

```
1  INSERT INTO airline_info_view
2      (airline_id, airline_code, airline_name, airline_country)
3  VALUES
4      (11, 'TEST', 'Test Airways', 'KAZAKHSTAN');
```

Data Output | Messages | Notifications

```
ERROR:  null value in column "created_at" of relation "airline" violates not-null constraint
Failing row contains (11, TEST, Test Airways, KAZAKHSTAN, null, null).

ОШИБКА:  null value in column "created_at" of relation "airline" violates not-null constraint
SQL state: 23502
Detail: Failing row contains (11, TEST, Test Airways, KAZAKHSTAN, null, null).
```

image(20261110-012210).png

Query | Query History | Scratch Pad

```
1  SELECT *
2  FROM airline_info_view
3  ORDER BY airline_id;
```

Data Output | Messages | Notifications

Showing rows: 1 to 11 | Page No: 1 of 1

	airline_id integer	airline_code character varying (30)	airline_name character varying (50)	airline_country character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone
1	1	KC	Air Astana	KAZAKHSTAN	2026-09-22 06:51:39.460902	2026-09-22 06:51:39.460902
2	2	FS	FlyArystan	KAZAKHSTAN	2026-09-22 06:51:39.460902	2026-09-22 06:51:39.460902
3	3	TK	Turkish Airlines	TURKEY	2026-09-22 06:51:39.460902	2026-09-22 06:51:39.460902
4	4	EK	Emirates	UNITED ARAB EMIRAT…	2026-09-22 06:51:39.460902	2026-09-22 06:51:39.460902
5	5	LH	Global Airways	United Kingdom	2026-09-22 06:51:39.460902	2026-09-28 21:00:35.318781
6	6	KZ	KazAir	TURKEY	2026-09-28 20:44:48.567733	2026-09-28 20:45:38.773551
7	7	AE	AirEasy	FRANCE	2026-09-28 20:46:19.547541	2026-09-28 20:46:19.547541
8	8	FH	Dana	BRAZIL	2026-09-28 20:46:19.547541	2026-09-28 20:46:19.547541
9	9	FF	FlyFly	POLAND	2026-09-28 20:46:19.547541	2026-09-28 20:46:19.547541
10	10	aa	wd	usze	2023-05-01 00:00:00	2026-05-02 00:00:00
11	11	TEST	Test Airways	KAZAKHSTAN	2026-11-10 06:23:43.568595	2026-11-10 06:23:43.568595

image(20261110-012415).png

Задание 10.

Query | Query History

```
1  SELECT table_name
2  FROM information_schema.views
3  WHERE table_schema = 'public'
4  ORDER BY table_name;
```

Data Output | Messages | Notifications

table_name name

image(20261110-012456).png

Часть II. Роли и права доступа (ROLES)

Задание 11.

Query | Query History

```sql
SELECT
    rolname,
    rolsuper,
    rolcreatedb,
    rolcreaterole,
    rolcanlogin
FROM pg_roles
WHERE rolname IN ('flight_manager', 'flight_analyst')
ORDER BY rolname;
```

Data Output | Messages | Notifications

Showing rows: 1 to 2

	rolname name	rolsuper boolean	rolcreatedb boolean	rolcreaterole boolean	rolcanlogin boolean
1	flight_analyst	false	false	false	false
2	flight_manag...	false	false	false	false

image(20261110-012605).png

Query | Query History

```sql
SELECT
    member.rolname AS username,
    role.rolname AS group_role
FROM pg_auth_members m
JOIN pg_roles role
    ON m.roleid = role.oid
JOIN pg_roles member
    ON m.member = member.oid
WHERE member.rolname IN ('first_user', 'second_user')
ORDER BY username;
```

Data Output | Messages | Notifications

Showing rows: 1 to 2

	username name	group_role name
1	first_user	flight_man…
2	second_us…	flight_analy…

image(20261110-012649).png

Задание 12.

Query | Query History

```sql
SELECT
    grantee,
    table_name,
    privilege_type
FROM information_schema.role_table_grants
WHERE grantee = 'flight_analyst'
  AND table_name IN ('flights', 'airline', 'booking')
ORDER BY table_name;
```

Data Output | Messages | Notifications

Showing rows: 1 to 3

	grantee name	table_name name	privilege_type character varying
1	flight_analy…	airline	SELECT
2	flight_analy…	booking	SELECT
3	flight_analy…	flights	SELECT

Successfully run.

image(20261110-012740).png

Задание 13.

Query | Query History

```sql
SELECT
    grantee,
    table_name,
    privilege_type
FROM information_schema.role_table_grants
WHERE grantee = 'flight_manager'
ORDER BY table_name, privilege_type;
```

Data Output | Messages | Notifications

Showing rows: 1 to 7

	grantee name	table_name name	privilege_type character varying
1	flight_manag...	airline	SELECT
2	flight_manag...	airport	SELECT
3	flight_manag...	booking	INSERT
4	flight_manag...	booking	SELECT
5	flight_manag...	booking	UPDATE
6	flight_manag...	flights	SELECT
7	flight_manag...	passengers	SELECT

image(20261110-012933).png

Задание 14.

Query | Query History

```
1  SELECT
2      grantee,
3      table_name,
4      privilege_type
5  FROM information_schema.role_table_grants
6  WHERE grantee = 'flight_manager'
7  ORDER BY table_name, privilege_type;
```

Data Output | Messages | Notifications

Showing rows: 1 to 7

	grantee name	table_name name	privilege_type character varying
1	flight_manag...	airline	SELECT
2	flight_manag...	airport	SELECT
3	flight_manag...	booking	INSERT
4	flight_manag...	booking	SELECT
5	flight_manag...	booking	UPDATE
6	flight_manag...	flights	SELECT
7	flight_manag...	passengers	SELECT

image(20261110-012933).png

Задание 15.

Query | Query History

```
1  SET ROLE second_user;
2
3  SELECT current_user;
```

Data Output | Messages | Notifications

Showing rows: 1 t

	current_user name
1	second_user

image(20261110-013428).png

Query | Query History

```
1  SELECT *
2  FROM flights;
```

Scratch Pad

Data Output | Messages | Notifications

Showing rows: 1 to 5 | Page No: 1 of 1

	flight_id [PK] integer	sch_departure_time timestamp without time zone	sch_arrival_time timestamp without time zone	departing_airport_id integer	arriving_airport_id integer	departing_gate text	arriving_gate character varying (50)	airline_id integer	act_departu timestamp
1	1	2026-09-25 08:00:00	2026-09-25 10:00:00	1	2	A1	B1	1	2026-09-25
2	2	2026-09-26 09:00:00	2026-09-26 13:00:00	2	3	A2	B2	3	2026-09-26
3	3	2026-09-27 12:00:00	2026-09-27 15:00:00	3	4	A3	B3	4	2026-09-27
4	4	2026-09-28 14:00:00	2026-09-28 18:00:00	4	5	A4	B4	5	2026-09-28
5	5	2026-09-29 16:00:00	2026-09-29 18:00:00	5	1	A5	B5	2	2026-09-29

image(20261110-013543).png

Query | Query History

```
1  DELETE FROM booking
2  WHERE booking_id = -1;
```

Data Output | Messages | Notifications

```
ERROR:  нет доступа к таблице booking

ОШИБКА:  нет доступа к таблице booking
SQL state: 42501
```

image(20261110-013612).png

Задание 16.

Query | Query History

```
1  SET ROLE first_user;
2
3  SELECT current_user;
```

Data Output | Messages | Notifications

	current_user name
1	first_user

image(20261110-013625).png

Query | Query History | Scratch Pad

```
SELECT *
FROM booking;
```

Data Output | Messages | Notifications

Showing rows: 1 to 6 | Page No: 1 of 1

	booking_id [PK] integer	flight_id integer	passenger_id integer	booking_platform character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone	status character varying (50)	ticket_price numeric (7,2)	ticket_discount numeric (10,2)
1	1	1	1	Website	2026-09-22 07:04:12.790704	2026-09-28 20:50:01.786866	Confirmed	172.50	[null]
2	2	2	2	Mobile App	2026-09-22 07:04:12.790704	2026-09-28 20:50:01.786866	Confirmed	322.00	[null]
3	3	3	3	Website	2026-09-22 07:04:12.790704	2026-09-28 20:50:01.786866	Confirmed	517.50	[null]
4	4	4	4	Travel Agency	2026-09-22 07:04:12.790704	2026-09-28 20:50:01.786866	Pending	598.00	[null]
5	5	5	5	Mobile App	2026-09-22 07:04:12.790704	2026-09-28 20:50:01.786866	Confirmed	218.50	[null]
6	1001	1	1	Online	2026-10-12 22:27:55.577641	2026-10-12 22:27:55.577641	Confirmed	10000.00	[null]

Successfully run. Total query runtime: 82 msec. 6 rows affected.

image(20261110-013736).png

Query | Query History | Scratch

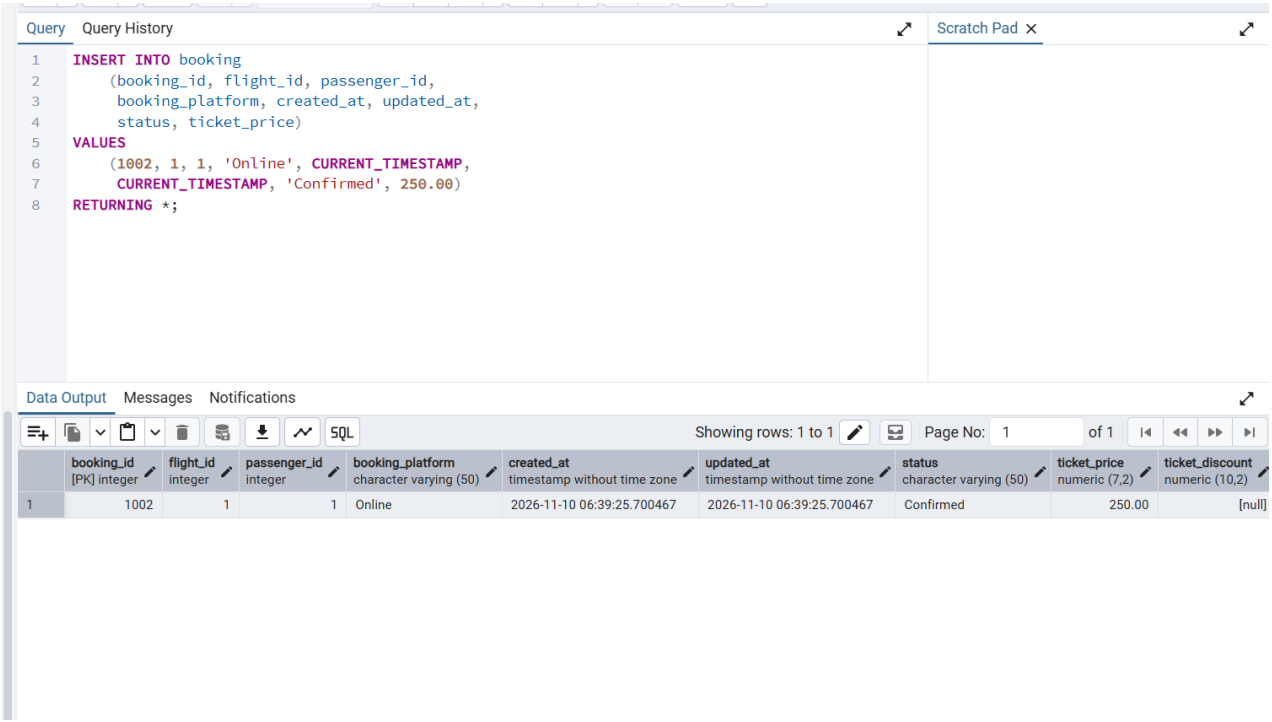
```
INSERT INTO booking
    (flight_id, passenger_id, booking_platform, status, ticket_price)
VALUES
    (1, 1, 'Online', 'Confirmed', 250.00)
RETURNING *;
```

Data Output | Messages | Notifications

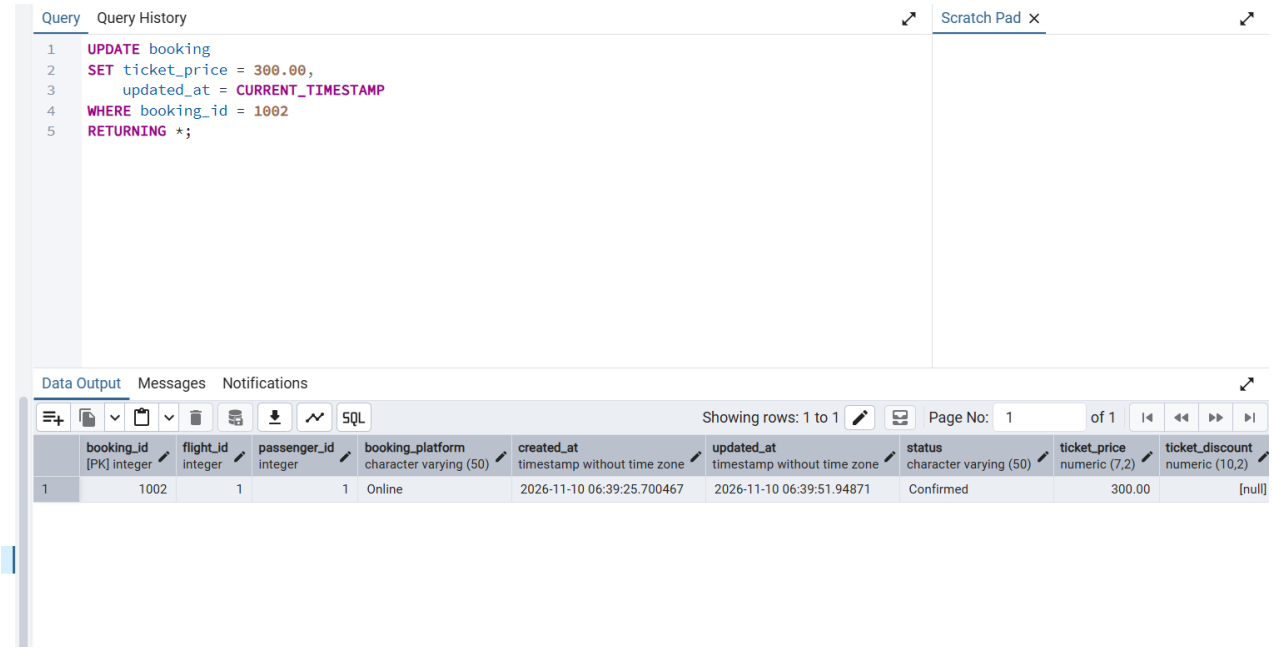
```
ERROR:  значение NULL в столбце "booking_id" отношения "booking" нарушает ограничение NOT NULL
Ошибочная строка содержит (null, 1, 1, Online, 2026-11-10 06:38:04.944462, null, Confirmed, 250.00, null).

ОШИБКА:  значение NULL в столбце "booking_id" отношения "booking" нарушает ограничение NOT NULL
SQL state: 23502
Detail: Ошибочная строка содержит (null, 1, 1, Online, 2026-11-10 06:38:04.944462, null, Confirmed, 250.00, null).
```

image(20261110-013826).png



image(20261110-013936).png



image(20261110-014004).png

Query | Query History

```
1  DELETE FROM booking
2  WHERE booking_id = 1002;
```

Data Output | Messages | Notifications

```
ERROR:  нет доступа к таблице booking

ОШИБКА:  нет доступа к таблице booking
SQL state: 42501
```

image(20261110-014124).png

Задание 17

Query | Query History

```
1  RESET ROLE;
2
3  SELECT current_user;
```

Data Output | Messages | Notifications

	current_user name
1	postgres

image(20261110-014357).png

Query Query History

Execute script F5

```
1  GRANT SELECT (first_name, last_name
2  ON passengers
3  TO flight_analyst;
```

Data Output Messages Notifications

```
GRANT

Query returned successfully in 52 msec.
```

image(20261110-014405).png

Query | Query History

Execute script F5

Scratch Pad

```
SELECT
    first_name,
    last_name,
    country_of_citizenship
FROM passengers;
```

Data Output | Messages | Notifications

Showing rows: 1 to 5 | Page No: 1

	first_name character varying (50)	last_name character varying (50)	country_of_citizenship character varying (50)
1	Leo	Messi	Kazakhstan
2	Pele	Maradona	Kazakhstan
3	Yamal	lamine	USA
4	Joan	Garcia	Spain
5	Gerd	Miller	Germany

Successfully run. Total query runtime: 77 mse

image(20261110-014425).png

Query Query History

Execute script
F5

```
1  SELECT *
2  FROM passengers;
```

Data Output Messages Notifications

```
ERROR:  нет доступа к таблице passengers

ОШИБКА:  нет доступа к таблице passengers
SQL state: 42501
```

image(20261110-014440).png

Query Query History

Execute script F5

```
1  SELECT first_name
2  FROM passengers;
```

Data Output Messages Notifications

Showing rows

	first_name character varying (50)
1	Leo
2	Pele
3	Yamal
4	Joan
5	Gerd

image(20261110-014510).png

Query Query History

```
1  SELECT passport_number
2  FROM passengers;
```

Data Output Messages Notifications

```
ERROR:  нет доступа к таблице passengers

ОШИБКА:  нет доступа к таблице passengers
SQL state: 42501
```

image(20261110-014522).png

Query | Query History

```
1  GRANT SELECT (first_name, last_name, country_of_citizenship)
2  ON passengers
3  TO flight_analyst;
```

Data Output | Messages | Notifications

```
GRANT

Query returned successfully in 91 msec.
```

image(20261110-014742).png

Query | Query History

```sql
SELECT
    grantee,
    table_name,
    column_name,
    privilege_type
FROM information_schema.role_column_grants
WHERE grantee = 'flight_analyst'
  AND table_name = 'passengers';
```

Data Output | Messages | Notifications

Showing rows: 1 to 3

	grantee name	table_name name	column_name name	privilege_type character varying
1	flight_analy...	passengers	country_of_citizens...	SELECT
2	flight_analy...	passengers	first_name	SELECT
3	flight_analy...	passengers	last_name	SELECT

image(20261110-014803).png

Задание 18.

Query | Query History

```
1  DELETE FROM booking
2  WHERE booking_id = -1;
```

Data Output | Messages | Notifications

```
ERROR:  нет доступа к таблице booking

ОШИБКА:  нет доступа к таблице booking
SQL state: 42501
```

image(20261110-013612).png

image(20261110-014440).png

image(20261110-014742).png

image(20261110-014829).png

Query Query History

```
1  GRANT USAGE, SELECT
2  ON SEQUENCE passengers_passenger_id_seq
3  TO flight_analyst;
```

Data Output Messages Notifications

```
ERROR:  отношение "passengers_passenger_id_seq" не существует

ОШИБКА:  отношение "passengers_passenger_id_seq" не существует
SQL state: 42P01
```

image(20261110-014843).png

Query | Query History

```
1  RESET ROLE;
2
3  SELECT column_name, data_type, is_nullable, column_default
4  FROM information_schema.columns
5  WHERE table_name = 'passengers'
6  ORDER BY ordinal_position;
```

Data Output | Messages | Notifications

Showing rows: 1 to 10

	column_name name	data_type character varying	is_nullable character varying (3)	column_default character varying
1	id1	integer	NO	[null]
2	first_name	character varying	NO	[null]
3	last_name	character varying	NO	[null]
4	date_of_birth	date	NO	[null]
5	gender	character varying	NO	[null]
6	country_of_citizens...	character varying	NO	[null]
7	country_of_residence	character varying	NO	[null]
8	passport_number	character varying	NO	[null]
9	created_at	timestamp without time zo...	NO	[null]
10	updated_at	timestamp without time zo...	NO	[null]

image(20261110-015104).png

Query | Query History

```
1  RESET ROLE;
2
3  SELECT id1, first_name, last_name
4  FROM passengers
5  ORDER BY id1;
```

Data Output | Messages | Notifications

Showing

	id1 [PK] integer	first_name character varying (50)	last_name character varying (50)
1	1	Leo	Messi
2	2	Pele	Maradona
3	3	Yamal	lamine
4	4	Joan	Garcia
5	5	Gerd	Miller
6	7	Test2	User

image(20261110-015451).png

image(20261110-015600).png